

Spectral Silence and Geometric Chaos in an Exactly Degenerate Topological Manifold

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Exact degeneracy silences every internal level-spacing diagnostic and elevates quantum geometry into the primary signal. We establish an independently testable hierarchy of projector correlations in a bosonic fractional-quantum-Hall zero-mode manifold. The energy spectral form factor obeys the exact identities $K_{E,\text{raw}} = D$ and $K_{E,c} = 0$, while the quantum-metric-normalized non-Abelian Berry curvature follows the finite- D complex-Jacobi ramp over a registered window. Structured-tangent and fixed-projector interventions resolve geometric chaos independently from active rank and intramultiplet spectral statistics; exact boundary atoms at $D > M$ yield the analytic connected plateau k/D . We then introduce a matrix-element Geometric-ETH test. On the genuine fixed-two-quasihole sequence $N = 3, 4, 5$, a covariance-whitened, gauge-invariant four-channel residual decreases to $R_4 = 0.20906$ and retains a resolved connected component relative to its matched Gaussian interval, selecting deformed Geometric ETH. On the closed twist torus, a smooth periodic ambient conjugation preserves the complete spectrum, gap, and Chern numbers $C_1 = 6, 10$ for $N = 3, 4$, while Wilson gap ratios change significantly and occupy a structured universality class distinct from circular-unitary transport. Exact degeneracy therefore reveals three complementary geometric layers: finite-Jacobi local correlations, operator-channel memory, and tunable non-Abelian holonomy inside one fixed topological class.

I. INTRODUCTION

Quantum chaos is conventionally diagnosed in the spectrum of a Hamiltonian. After resolving symmetries and unfolding the density of states, chaotic levels repel and their long-range correlations approach a Wigner–Dyson random-matrix ensemble [1, 2]. The connected spectral form factor packages the same two-level correlations into a ramp and plateau. The eigenstate thermalization hypothesis (ETH) complements this spectral picture by describing matrix elements of observables in the energy basis [3–5]. All three constructions use nonzero energy differences, so exact degeneracy opens a complementary regime in which state-space geometry carries the resolving information. For D states sharing the energy E_0 ,

$$\begin{aligned} Z_E(t) &= \sum_{a=1}^D e^{-itE_a} = D e^{-itE_0}, \\ K_{E,\text{raw}}(t) &= \frac{|Z_E(t)|^2}{D} = D, \end{aligned} \quad (1)$$

and the ensemble-connected form factor is identically zero. These identities exhaust the internal energy statistic while preserving the full projector and its response over parameter space.

The alternative studied here is geometric. An isolated degenerate eigenspace defines a vector bundle over Hamiltonian parameter space. Its adiabatic transport is governed by the Wilczek–Zee connection and the matrix-valued Berry curvature [6–8]; the symmetric partner is the quantum metric [9, 10]. Adiabatic eigenstate deformations are already known to be sensitive probes of integrability breaking [11], and the geometry of parameter space can distinguish ergodic from integrable random-

matrix regimes [12]. Most directly, Chen, Colin-Ellerin, Mamroud, and Papadodimas proposed non-Abelian Berry curvature as an intrinsic chaos diagnostic for exactly degenerate BPS microstates [13]. Their contrast is between structured, often vanishing curvature for smooth horizonless sectors and random-matrix-like curvature for supersymmetric black hole microstates.

We test that organizing principle in a purely condensed-matter fractional topological manifold generated by a local frustration-free parent. The manifold is the kernel of a positive semidefinite bosonic fractional-quantum-Hall parent Hamiltonian. Its degeneracy is exact and algebraic, and its projector is separated from the complement by a gap. This setting isolates the proposed geometric diagnostic from the special BPS mechanism and makes five questions answerable:

1. Does a curvature ramp survive when the energy form factor is exactly silent?
2. Does that ramp distinguish random local tangents from structured tangents at the same active rank?
3. Can energy chaos and projector-geometric chaos be varied independently?
4. Do gauge-invariant response-channel matrix elements approach a covariance-matched Wick law along a genuine many-body sequence?
5. Can non-Abelian holonomy change inside a closed bundle while its complete spectrum and Chern class remain fixed?

Our answers establish five complementary advances across spectrum, local geometry, response tensors, topology, and finite-rank structure. The metric-normalized

curvature is a compression of a fixed signature matrix by the row space of projector-to-complement response channels. A Haar row space gives an exact finite-dimensional complex Jacobi point process, allowing us to derive its connected form factor directly at the finite rank used in the calculation. The physical random-local-tangent ensemble follows this prediction over the registered two-level window. A complete momentum-resolved tangent family supplies a same-rank structured reference with only a small number of distinct curvature eigenvalues. The adjacent-gap ratio, connected form factor, and number variance place this reference in a clearly resolved structured regime.

The central evidence is causal rather than correlational. We vary the Hamiltonian restricted to the degenerate fiber, PHP , from Poisson to GUE statistics while holding P and the projector-to-complement channels fixed; the energy spectrum changes, whereas the Berry curvature remains exactly invariant. Separately, we interpolate the tangent channels from Fourier structured to random local directions at fixed parent Hamiltonian. Local Jacobi repulsion appears at a smaller interpolation strength than the full registered form-factor window, while long-range number variance retains microscopic memory. The resulting order

$$\begin{aligned} \text{local repulsion} &\longrightarrow \text{two-level ramp} \\ &\longrightarrow \text{long-range rigidity} \end{aligned} \quad (2)$$

is a measured hierarchy of geometric correlations. The form-factor variable is a dimensionless Fourier scale of unfolded curvature eigenvalues. Its connection to real-time dynamics defines a separate and promising extension.

Beyond eigenvalue correlations, fixed local operators expose a resolved four-channel response cumulant. Its normalized residual decreases from $N = 3$ to $N = 5$, demonstrating progressive Gaussianization together with finite-size operator memory relative to the covariance-matched reference. On the closed twist torus, a globally periodic ambient conjugation leaves every energy and the first Chern class invariant while changing Wilson-loop statistics reproducibly. The resulting Wilson statistics occupy a structured regime distinct from CUE. Together, the response tensor and Wilson sector map how local Jacobi behavior coexists with higher and global microscopic structure.

Finally, an independent rank sequence through $D = 800$ resolves a finite-dimensional effect unique to signature-compressed ensembles. When the active rank exceeds the dimension M of either signature polarization, exact atoms appear at $\lambda = \pm 1$. We prove that these deterministic modes reduce the connected plateau from one to k/D , where $k = 2M - D$ is the continuous-sector dimension, while the continuous complement retains the Jacobi ramp. This resolves exact geometric modes and correlated continuous modes within one curvature matrix.

The paper is organized as follows. Section II derives the projector response, signature compression, finite-Jacobi kernel, and atom theorem. Section III introduces the

fractional topological manifold and the structured same-rank comparison. Section IV performs the independent spectral and geometric interventions. Section V gives the finite-rank atom form factors. Section VI resolves the correlation hierarchy and its statistical meaning. Section VII tests response-channel Wick factorization and fixed-Chern Wilson holonomy. Appendix A records the numerical and inference protocols. Appendix B gives the gauge-covariance, Wick, Gram-reduction, and bundle-isomorphism derivations.

II. PROJECTOR GEOMETRY AND FINITE JACOBI THEORY

A. Gauge-invariant response of a degenerate projector

Let $H(\lambda)$ have an isolated D -dimensional eigenspace at energy E_0 , with spectral projector P , complement $Q = 1 - P$, and reduced resolvent

$$R = Q[Q(H - E_0)Q]^{-1}Q. \quad (3)$$

For the tangent $V_\mu = \partial_\mu H$, define the projector-to-complement response channel

$$X_\mu = PV_\mu R. \quad (4)$$

In bases of the degenerate fiber and its coupled complement, X_μ is a $D \times M$ matrix. Kato perturbation theory gives, for two real directions v, w ,

$$F_{vw} = i(X_v X_w^\dagger - X_w X_v^\dagger), \quad \Gamma = X_v X_v^\dagger + X_w X_w^\dagger. \quad (5)$$

Here F_{vw} is Hermitian under our convention, and Γ is the positive two-direction quantum metric. Both transform by conjugation under a basis change within P , so the generalized eigenvalues of (F_{vw}, Γ) are gauge invariant.

The raw curvature scale depends on tangent normalization and energy denominators. We remove it by restricting to the $r = \text{rank } \Gamma$ support and defining

$$\Omega = \Gamma^{-1/2} F_{vw} \Gamma^{-1/2}. \quad (6)$$

This normalization is algebraic and model agnostic. It converts the diagnostic into the relative orientation of the two response channels. With

$$B = (X_v \ X_w), \quad J = \begin{pmatrix} 0 & iI_M \\ -iI_M & 0 \end{pmatrix}, \quad (7)$$

whitening gives

$$Y = \Gamma^{-1/2} B, \quad YY^\dagger = I_r, \quad \Omega = Y J Y^\dagger. \quad (8)$$

Thus Ω is a compression of a fixed matrix with M positive and M negative eigenvalues. Its statistics depend only on an r -plane in $\mathbb{C}^{2M}$, i.e., a point on a complex Grassmannian. This observation makes the random-matrix reference precise: randomness is assigned exclusively to the orientation of the whitened response row space.

B. Finite complex-Jacobi point process

If the row space of Y is Haar distributed, the continuously distributed eigenvalues of Eq. (8) form a complex Jacobi ensemble [14, 15]. For $r \leq M$, all $k = r$ levels lie in $(-1, 1)$ and

$$p(\lambda_1, \dots, \lambda_k) \propto \prod_{a < b} |\lambda_a - \lambda_b|^2 \prod_{a=1}^k (1 - \lambda_a^2)^{M-r}. \quad (9)$$

More generally, let

$$k = \min(r, 2M - r), \quad a = |M - r|. \quad (10)$$

After the exact boundary levels derived below are removed, the interior joint density has the same form as Eq. (9) with r replaced by k and exponent a .

Let $p_n^{(a)}(x)$ be orthonormal on $(-1, 1)$ with weight $w_a(x) = (1 - x^2)^a$, and define the weighted projection kernel

$$\mathcal{K}_k(x, y) = \sqrt{w_a(x)w_a(y)} \sum_{n=0}^{k-1} p_n^{(a)}(x)p_n^{(a)}(y). \quad (11)$$

Its diagonal $\rho_k(x) = \mathcal{K}_k(x, x)$ integrates to k . The finite-rank unfolding is therefore

$$u(x) = \int_{-1}^x \rho_k(s) ds. \quad (12)$$

For the unfolded partition sum $Z_F(\tau) = \sum_{j=1}^k e^{-2\pi i \tau u(\lambda_j)}$, we use one normalization throughout:

$$\begin{aligned} K_{F,\text{raw}}(\tau) &= \frac{1}{k} \langle |Z_F(\tau)|^2 \rangle, \\ K_{F,\text{disc}}(\tau) &= \frac{1}{k} |\langle Z_F(\tau) \rangle|^2, \\ K_{F,c}(\tau) &= K_{F,\text{raw}}(\tau) - K_{F,\text{disc}}(\tau). \end{aligned} \quad (13)$$

The determinantal identity $\rho_2(x, y) = \rho_k(x)\rho_k(y) - |\mathcal{K}_k(x, y)|^2$ then yields the exact finite- k prediction

$$K_{J,c}^{(k)}(\tau) = 1 - \frac{1}{k} \int_{-1}^1 dx \int_{-1}^1 dy e^{-2\pi i \tau [u(x) - u(y)]} |\mathcal{K}_k(x, y)|^2. \quad (14)$$

Equation (14) generates the finite- D ramp directly and gives $K_{J,c}^{(k)}(0) = 0$ from the projector identity and approaches the unit plateau when the oscillatory cluster term is washed out. We evaluate the double integral by Gauss-Jacobi quadrature; convergence and an independent matrix-beta Monte Carlo comparison are given in Appendix A.

The word “form factor” in Eq. (13) denotes a Fourier transform of the curvature point process, with τ serving as a dimensionless geometric Fourier coordinate. A future dynamical extension can connect this coordinate to operator growth and thermalization scales.

C. Boundary-atom theorem and form-factor plateau

When $r > M$, the row space $\mathcal{S} \subset \mathbb{C}^{2M}$ must intersect both M -dimensional eigenspaces $\mathcal{H}_{\pm}$ of J :

$$\dim(\mathcal{S} \cap \mathcal{H}_{\pm}) \geq r + M - 2M = r - M. \quad (15)$$

Every vector in these intersections is an exact eigenvector of the compression with eigenvalue ± 1 . For a generic r -plane the bound is saturated, giving

$$m_+ = m_- = r - M, \quad k = 2M - r. \quad (16)$$

The atoms are deterministic under the Jacobi ensemble, so their connected covariance is exactly zero. If the full spectrum is normalized by its total rank r , the exact connected form factor is consequently

$$K_{J,c}^{\text{full}}(\tau) = \frac{k}{r} K_{J,c}^{(k)}(\tau), \quad K_{J,c}^{\text{full}}(\infty) = \frac{k}{r}. \quad (17)$$

This is an exact finite-dimensional plateau theorem.

For completeness, the raw, non-unfolded form factor retains deterministic interference with the atoms. Writing

$$\begin{aligned} Z_{\text{full}}(s) &= Z_{\text{at}}(s) + Z_{\text{int}}(s), \\ Z_{\text{at}}(s) &= m_- e^{2\pi i s} + m_+ e^{-2\pi i s}, \end{aligned} \quad (18)$$

gives the exact decomposition

$$K_{\text{raw}}^{\text{full}} = \frac{|Z_{\text{at}}|^2}{r} + \frac{2 \text{Re}[Z_{\text{at}}^* \langle Z_{\text{int}} \rangle]}{r} + \frac{\langle |Z_{\text{int}}|^2 \rangle}{r}. \quad (19)$$

Equations (17) and (19) distinguish deterministic geometry from correlated fluctuations and provide two independent numerical closure tests.

III. ENERGY-SILENT FRACTIONAL TOPOLOGICAL MANIFOLD

A. Exact zero modes and tangent families

Our physical anchor is a bosonic Kapit-Mueller lattice parent projected to its exactly flat lowest band [16]. At particle number $N = 3$ and flux number $n = 10$, the projected contact interaction is positive semidefinite and has an isolated zero-mode manifold of dimension $D = 50$. The complement reached by local potentials has dimension $M = 170$ per tangent direction. The zero modes are lattice Laughlin quasihole states obeying the cyclic $(1, 2)$ exclusion rule [17–19]. The numerical kernel width is 5.78×10^{-16} , whereas the external gap is 0.106672.

This is an exact flat band generated by a purely condensed-matter, frustration-free parent construction. Its kernel is algebraic and provides a mechanism complementary to supercharge, index, and BPS-shortening mechanisms. The open external gap makes P , R , and F_{vw} well defined, while the exact internal degeneracy enforces Eq. (1).

We compare two local-potential tangent families at this same parent:

$$V[v] = \sum_{\mathbf{x}} v_{\mathbf{x}} \hat{n}_{\mathbf{x}}, \quad \sum_{\mathbf{x}} v_{\mathbf{x}} = 0. \quad (20)$$

The physical ensemble contains 20000 independent random tangent pairs, divided before analysis into 12000/4000/4000 training, validation, and test matrices. Every result below uses the untouched 4000-matrix test set.

The structured reference is the complete nonzero Fourier family on the 5×5 lattice. For momentum $\mathbf{q} \neq 0$, take

$$v_{\mathbf{x}}^{(\mathbf{q})} \propto \cos(\mathbf{q} \cdot \mathbf{x}), \quad w_{\mathbf{x}}^{(\mathbf{q})} \propto \sin(\mathbf{q} \cdot \mathbf{x}), \quad (21)$$

normalized in the same many-body tangent metric as the random potentials. All 24 nonzero momenta are enumerated, but $\mathbf{q}$ and $-\mathbf{q}$ combine into 12 independent orbit-level statistical units. Each Fourier pair has the same active rank $r = 50$ as a random pair, yet its curvature has at most ten distinct eigenvalues. It therefore separates tangent structure from rank, gap, parent Hamiltonian, and normalization.

B. Spectral silence and a geometric ramp

Figure 1 states the main contrast. The exact energy form factor follows Eq. (1): the raw curve is $D = 50$, and the connected curve is 0 for every t . The curvature spectrum, however, distinguishes structured Fourier tangents from random local tangents at the same rank.

At $\tau = 0.5$, the connected curvature form factors are

$$\begin{aligned} K_{F,c}^{\text{structured}} &= 4.476, \\ K_{F,c}^{\text{physical}} &= 0.502, \quad K_{J,c}^{(50)} = 0.501. \end{aligned} \quad (22)$$

The physical value is thus close to the parameter-free finite-rank prediction, whereas the structured value is almost an order of magnitude larger. The agreement is assessed on the whole registered interval with simultaneous bands; Eq. (22) provides one representative value.

C. Three-scale comparison

A robust geometric-chaos diagnostic combines three resolutions of the same point process:

$$P(r) \longleftrightarrow K_{F,c}(\tau) \longleftrightarrow \Sigma^2(L), \quad (23)$$

testing nearest-neighbor repulsion, two-level correlations across Fourier scales, and long-range counting fluctuations, respectively. Related uses of non-energy spectra emphasize that a ramp probes eigenvalue correlations rather than the physical identity of the operator [20].

Figure 2 resolves two distinct geometric regimes. First, the rank-50 Fourier reference occupies the structured branch across all three tests. Second, the physical curvature combines Jacobi local statistics with covariance-deformed structure at broader scales. Its one-point density has held-out L^1 distance 0.2910 from Haar; a covariance-deformed row-space model reduces this to 0.1237, a 57.5% improvement. At long range,

$$\Sigma_{\text{phys}}^2(8) - \Sigma_{\text{Haar}}^2(8) = 0.12582 \in [0.11428, 0.13736]. \quad (24)$$

The resulting statement is correlation-scale resolved: Jacobi local repulsion and a finite- D ramp coexist with covariance-deformed one-point structure and long-range memory.

IV. INDEPENDENT SPECTRAL AND GEOMETRIC INTERVENTIONS

The preceding comparison could still admit a weak interpretation: perhaps the energy and curvature statistics are merely two correlated consequences of a single hidden randomness. We now intervene on the two blocks of a parameter-dependent Hamiltonian separately. The decomposition

$$H = PHP + PHQ + QHP + QHQ \quad (25)$$

shows why this is possible. Internal energy spacings are controlled by PHP , whereas the projector response in Eq. (4) is controlled by $P(\partial_{\mu}H)Q$, the complement resolvent, and their adjoints. These are logically distinct data.

A. Changing energy statistics at fixed projector

For the spectral intervention we use an effective $D \times D$ fiber Hamiltonian

$$h_{\alpha} = \sqrt{1 - \alpha^2} h_P + \alpha h_{\text{GUE}}, \quad 0 \leq \alpha \leq 1, \quad (26)$$

where h_P has independent exponential spacings and h_{GUE} is an independent complex Gaussian Hermitian matrix. Both are centered and matched in Hilbert–Schmidt norm before interpolation. We embed $0.1h_{\alpha}$ in a fixed target block and place a diagonal complement between energies 2 and 3. The external gap stays open throughout, and diagonalization verifies

$$\max_{\alpha} \|P_{\alpha} - P_0\|_F = 1.15 \times 10^{-14}. \quad (27)$$

The tangent channels and a reference curvature spectrum are held fixed. This effective causal control supplies a clean spectral axis; a local microscopic realization defines a complementary future program.

Across 4000 matrices per value of α , the energy adjacent-gap ratio changes from

$$\langle r_E \rangle_{\alpha=0} = 0.385645, \quad \langle r_E \rangle_{\alpha=1} = 0.598677, \quad (28)$$

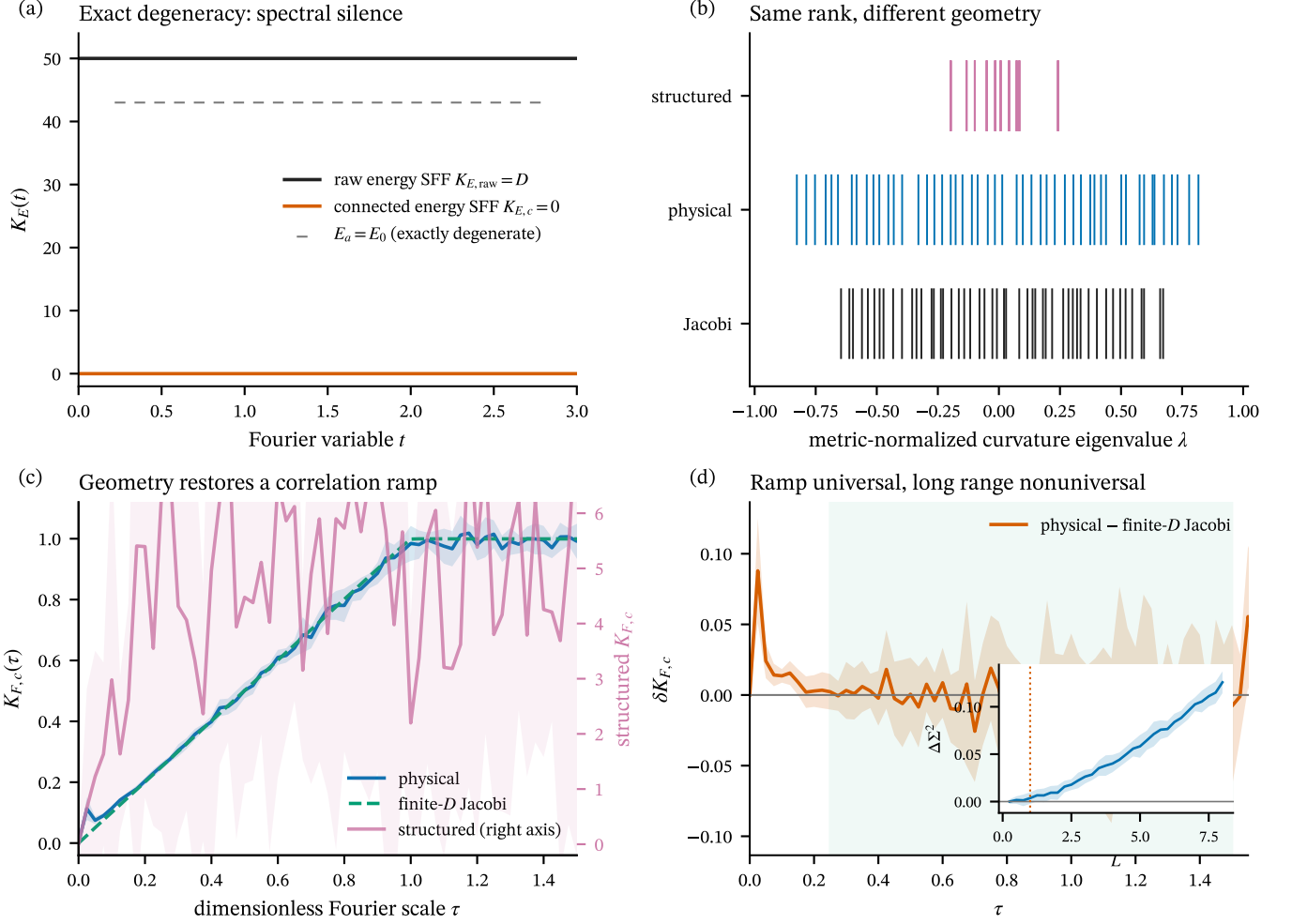


FIG. 1. **Exact degeneracy silences the energy spectrum, while projector geometry retains correlated fluctuations.** (a) For $D = 50$ exactly degenerate zero modes, the raw energy SFF is the featureless constant D , and the connected SFF vanishes identically. The marks display the coincident energies of the exact manifold. (b) Representative spectra of the metric-normalized curvature Ω : structured Fourier tangents retain multiplets, whereas random local tangents produce a resolved spectrum comparable to the Haar–Jacobi reference. (c) Connected curvature SFF of the 4000-matrix physical test ensemble and the exact finite- D Jacobi prediction from Eq. (14). The Fourier control is plotted against a separate right axis because it is parametrically larger. (d) The physical-minus-Jacobi residual has a simultaneous confidence band compatible with zero over the registered interval $0.25 \leq \tau \leq 1.5$, while the inset shows that physical-minus-Haar number variance becomes positive beyond $L \simeq 1$. Here t in (a) is conjugate to energy, whereas τ in (c,d) is only a dimensionless Fourier variable for unfolded curvature eigenvalues.

while the stored curvature spectrum is unchanged to machine precision. Thus conventional spectral chaos evolves at exactly fixed target-projector geometry.

B. Changing geometry at fixed exactly degenerate parent

The complementary intervention leaves the Kapit–Mueller parent, its exact zero energy, its projector, and its external gap fixed. For each structured Fourier direction

$v_{\mathbf{q}}$ and an independent random local direction η_v , define

$$\begin{aligned} \tilde{v}_g &= \mathcal{N}_G \left[\sqrt{1-g} v_{\mathbf{q}} + \sqrt{g} \eta_v \right], \\ \tilde{w}_g &= \mathcal{N}_G \left[\sqrt{1-g} w_{\mathbf{q}} + \sqrt{g} \eta_w \right], \end{aligned} \quad (29)$$

where $\mathcal{N}_G$ mean-centers and normalizes in the physical tangent Gram matrix. The square-root weights make g a variance fraction before the final normalization. We sample 4000 independent tangent pairs at each positive registered g , maintaining active rank 50 throughout.

Figure 3 gives the operational separation. Changing *PHP* moves horizontally in panel (d) at fixed $P(\partial H)Q$. Changing local tangent structure moves vertically at fixed

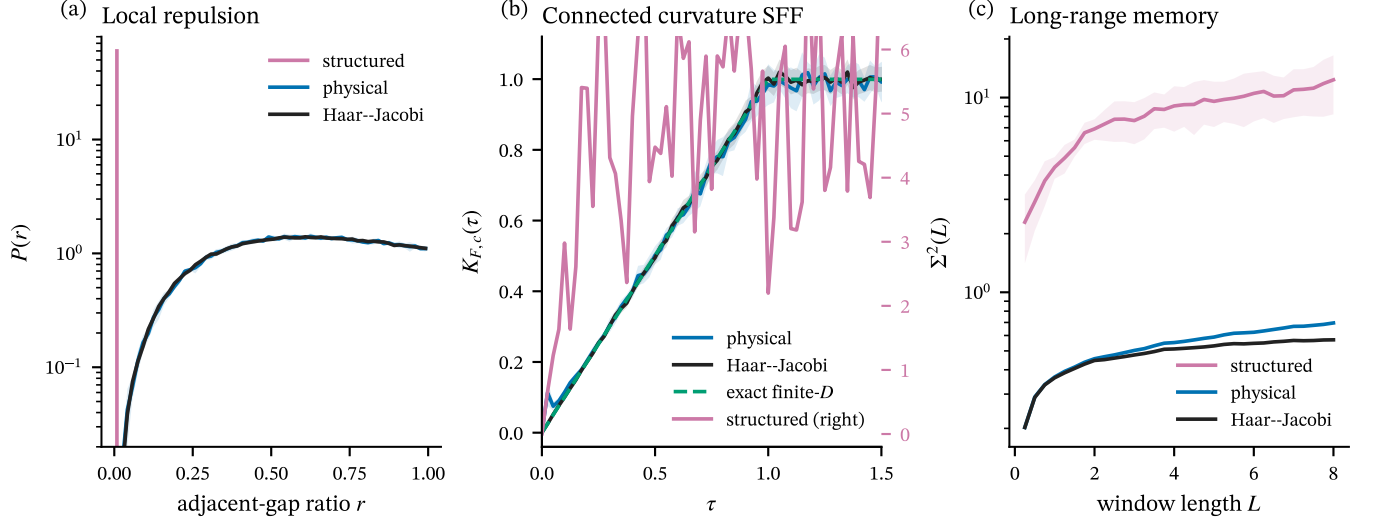


FIG. 2. **Three-scale comparison for geometric chaos.** (a) Adjacent-gap-ratio distribution of structured Fourier, physical random-local, and independently sampled Haar–Jacobi curvatures. Exact multiplets make the structured distribution singular near zero; the physical and Jacobi bands overlap. (b) Connected curvature SFF. The empirical Haar curve and the analytic determinantal-kernel result are independent references. (c) Number variance after ensemble-CDF unfolding. Random local tangents have Jacobi rigidity at short windows but a resolved excess at long windows, whereas the structured control is nonuniversal throughout. Bands are simultaneous 95% regions obtained from complete momentum orbits, physical seed blocks, or complete random matrices, with each matrix retained as one statistical unit.

exactly degenerate parent. Consequently, a Jacobi-like curvature spectrum directly diagnoses the embedding and response of the degenerate subspace. It diagnoses how the degenerate subspace is embedded in and responds to its surrounding Hilbert space.

The intervention also supplies an ordering test. The mean curvature gap-ratio interval first overlaps the Haar reference at $g = 0.20$, whereas compatibility with the exact finite- D SFF over a registered nonzero τ interval first occurs at $g = 0.40$. These onsets are confidence-set definitions on a discrete grid. They demonstrate that local repulsion is established before the broader two-level ramp under controlled tangent interpolation.

V. EXACT GEOMETRIC MODES AND THE JACOBI PLATEAU

The $D = 50$ lattice calculation lies in the atom-free regime $D < M$. To test Eq. (17), we use an independent exact root-response construction that reaches both sides of the capacity boundary. For $N = 3$ particles in n orbitals, the cyclic Laughlin zero-mode dimension and one-tangent descendant capacity are

$$D(n) = \frac{n}{n-3} \binom{n-3}{3}, \quad M(n) = 2n(n-3). \quad (30)$$

Two independently sampled one-body response maps are assembled in the monomial root basis and whitened by a thin QR factorization. This directly produces the row

isometry in Eq. (8) as an independent construction from the physical Kapit–Mueller calculation.

The registered sequence contains 8750 matrices at

$$D = 16, 50, 112, 210, 352, 546, 800.$$

It is a finite-rank sequence at fixed particle number $N = 3$, designed to resolve the exact capacity transition. At $D = 546$, $M = 540$ and the atom theorem requires six levels at each boundary. At the endpoint $(D, M) = (800, 680)$, it requires 120 levels at each boundary and leaves $k = 560$ continuously distributed levels.

The endpoint plateau is therefore fixed analytically:

$$K_{F,c}^{\text{full}}(\infty) = \frac{2M - D}{D} = 0.7. \quad (31)$$

The reference curve is obtained in two independent ways: evaluation of the determinantal kernel in Eq. (14) and matrix-beta sampling of the Jacobi ensemble. The response spectra are generated independently from both references.

This result refines the meaning of random-matrix-like curvature. The full one-point measure at $D > M$ contains a finite mass fixed at ± 1 , while the atom-stripped sector displays Jacobi repulsion and its finite-rank ramp. The exact modes record a dimension-theorem intersection of the accessible row space with the two polarizations of J ; the ramp probes correlations in the remaining orientation. Exact geometry and chaotic geometry are therefore compatible within orthogonal sectors of the same degenerate manifold.

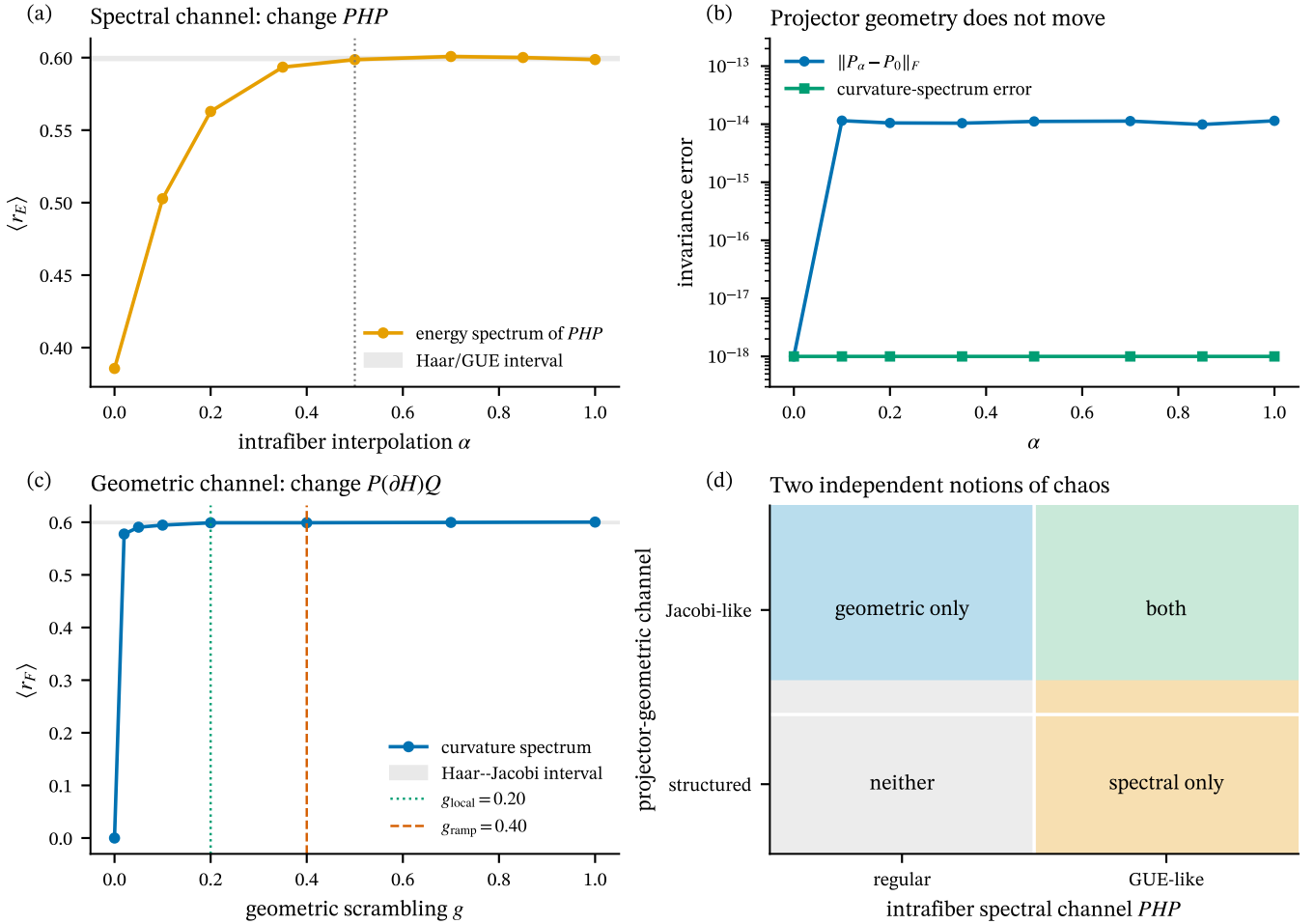


FIG. 3. **Energy chaos and projector-geometric chaos are independent channels.** (a) Mean energy adjacent-gap ratio under the effective fixed-projector interpolation in Eq. (26); it crosses from Poisson to the GUE/Haar interval. (b) The target-projector distance remains at diagonalization roundoff and the supplied curvature spectrum is exactly invariant. (c) At fixed exactly degenerate parent, scrambling only the tangent channels in Eq. (29) changes the curvature gap ratio. Its simultaneous interval first overlaps the Haar-Jacobi interval at $g = 0.20$; the first registered nontrivial SFF window occurs at $g = 0.40$. (d) Logical 2×2 classification. Spectral and geometric chaos can occur separately or together; the original topological flat band occupies the energy-silent/geometrically chaotic quadrant. Bands are 95% simultaneous intervals over the registered interpolation grids.

The algebraic atom count also strengthens the analysis. Counts are fixed from (D, M) before eigenvalues are examined, their numerical positions are audited afterward, and both full and continuous-sector normalizations are reported.

VI. A HIERARCHY OF GEOMETRIC CORRELATION SCALES

The intervention data allow a scale-resolved statement stronger than a single endpoint comparison. For each g , we subtract the exact finite- D Jacobi curve and test the entire residual process with a simultaneous band. For the physical ensemble we repeat the same procedure using eight independently seeded blocks. Long-range rigidity is

tested independently through the number variance

$$\Sigma^2(L) = \left\langle [N_\xi(x, L) - L]^2 \right\rangle, \quad (32)$$

where $N_\xi(x, L)$ counts unfolded curvature levels in a sliding interval of length L .

Three conclusions follow from Fig. 5. First, a small interpolation away from Fourier structure generates local level repulsion, while the two-level correlation curve becomes active at a larger interpolation strength. The registered local and ramp onsets are

$$g_{\text{local}} = 0.20, \quad g_{\text{ramp}} = 0.40. \quad (33)$$

Second, at the physical random-local endpoint the residual band contains zero for every registered $\tau \in [0.25, 1.5]$.

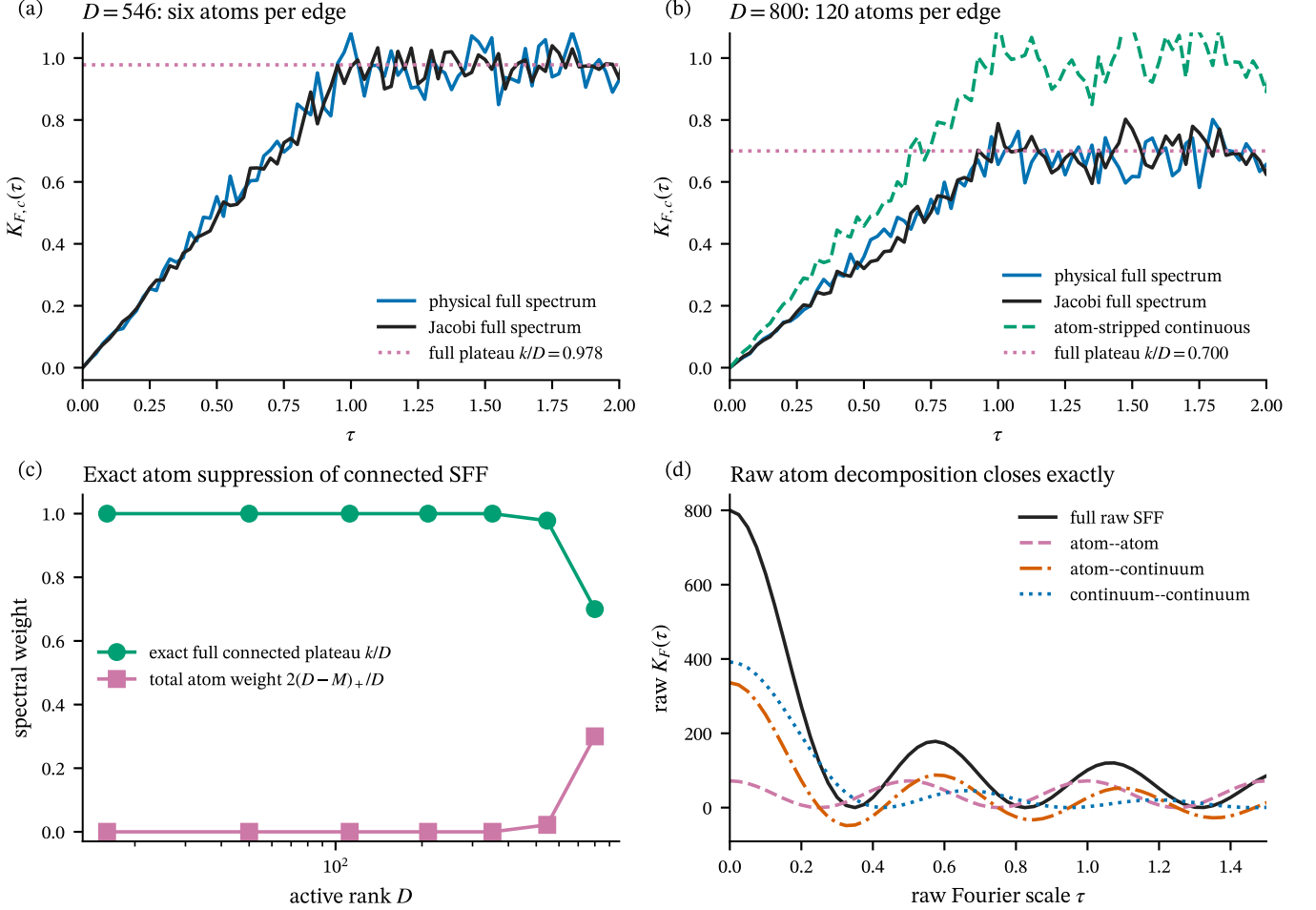


FIG. 4. **Exact boundary atoms reshape the connected plateau while the continuous sector retains the Jacobi ramp.** (a) Full-spectrum connected curvature SFF at the first capacity crossing, $D = 546$, $M = 540$, with six exact levels at each of $\lambda = \pm 1$. (b) Endpoint $D = 800$, $M = 680$, with 120 atoms per boundary. The full physical root-response curve follows the full Jacobi prediction, while removing the algebraically identified atoms restores the unit continuous-sector plateau. (c) Across the seven-rank sequence, the exact full connected plateau is k/D ; it falls as the deterministic atom weight $2(D - M)_+/D$ grows. (d) Raw, non-unfolded endpoint SFF separated into atom-atom, atom-continuum, and continuum-continuum terms according to Eq. (19). Their sum closes the full raw form factor to 1.14×10^{-13} .

This is a curve-level finite- D test; it is more restrictive than matching the value or slope at one point.

Third, long-range rigidity defines an additional scale. The physical number variance follows the Haar reference through $L = 1.0$, and at $L = 8$ Eq. (24) resolves a positive simultaneous interval. The curvature point process therefore loses information in the ordered sequence

$$\begin{aligned} \text{local } P(r) &\prec \text{finite-}D \text{ SFF ramp} \\ &\prec \text{long-range } \Sigma^2(L) \end{aligned} \quad (34)$$

where $A \prec B$ means that A becomes compatible with the invariant reference at a weaker scrambling or shorter correlation scale.

Equation (34) defines a geometric Thouless hierarchy in the precise sense of confidence-resolved correlation lengths in an unfolded curvature spectrum. Mapping

these geometric scales to transport, scrambling, and real-time diffusion is a promising dynamical extension.

Our geometric-ETH statement is operational and complementary to energy-resolved ETH: after removal of a deterministic metric envelope, local and intermediate-range correlations of the projector-response spectrum approach those of a finite invariant point process. The held-out one-point covariance improvement and the surviving number variance show that higher Grassmannian information remains measurable. The result is covariance-deformed geometric ergodicity.

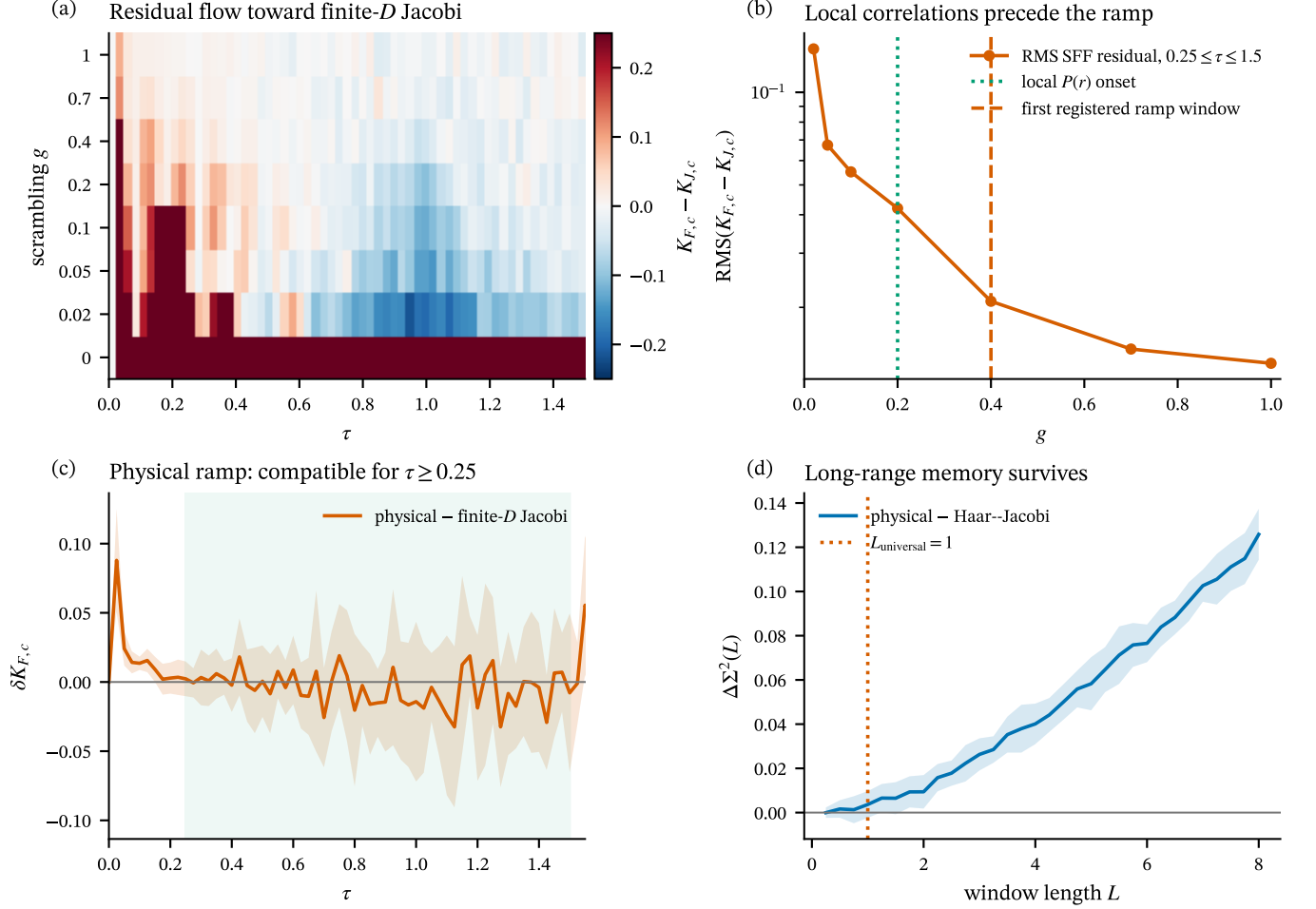


FIG. 5. **Controlled hierarchy from structured to random projector geometry.** (a) Residual connected SFF $K_{F,c}(\tau; g) - K_{J,c}^{(50)}(\tau)$ along the tangent-scrambling interpolation. (b) Root-mean-square residual over the preregistered range $0.25 \leq \tau \leq 1.5$. The local adjacent-gap statistic reaches the Haar interval at $g = 0.20$, while the first nontrivial registered SFF compatibility window occurs at $g = 0.40$. (c) Physical random-local-tangent residual with a simultaneous 95% band; it is compatible with the exact finite- D curve throughout the registered range. (d) Physical-minus-Haar number variance. Short windows are compatible through $L = 1.0$, but the positive long-range residual is resolved thereafter. Onset lines summarize confidence sets on the sampled grids.

VII. MATRIX-ELEMENT GEOMETRY AND FIXED-CHERN HOLONOMY

The curvature spectrum establishes a geometric correlation hierarchy. We now resolve its matrix-element mechanism and its relation to the topology of the full bundle while preserving the exact zero modes.

A. A gauge-invariant matrix-element test

Fixed local operators supply an independent test of Gaussian response statistics, because Gaussian tangent coefficients would make $X[V]$ Gaussian by linearity. We use eight simple mean-zero local density operators in each panel and form the resolvent-dressed maps $X_\mu = R_Q Q V_\mu P$. After whitening their measured label covari-

ance, the primary observable is the invariant tensor in Eq. (B4). Its two covariance-matched Gaussian contractions are derived in Appendix B; generalized ETH likewise elevates higher matrix-element correlations beyond entry-wise statistics [21, 22].

The finite-size reference is a distribution of R_4 at matched (D, M, m) . For every physical panel we generate Gaussian response tensors with the measured target and external covariance spectra, whiten them by the same algorithm, and compare the normalized residual in Eq. (B9). Complete panels are the resampling units. Independent target- and complement-frame rotations change the raw entries while T , T^{Wick} , and R_4 remain invariant.

Figure 6 uses the true sequence $n_\phi = 2N + 2$. The Hilbert space reaches 4368 states and the zero-mode rank reaches $D = 36$. The median physical residual decreases from 0.37093 to 0.24715 to 0.20906, while the

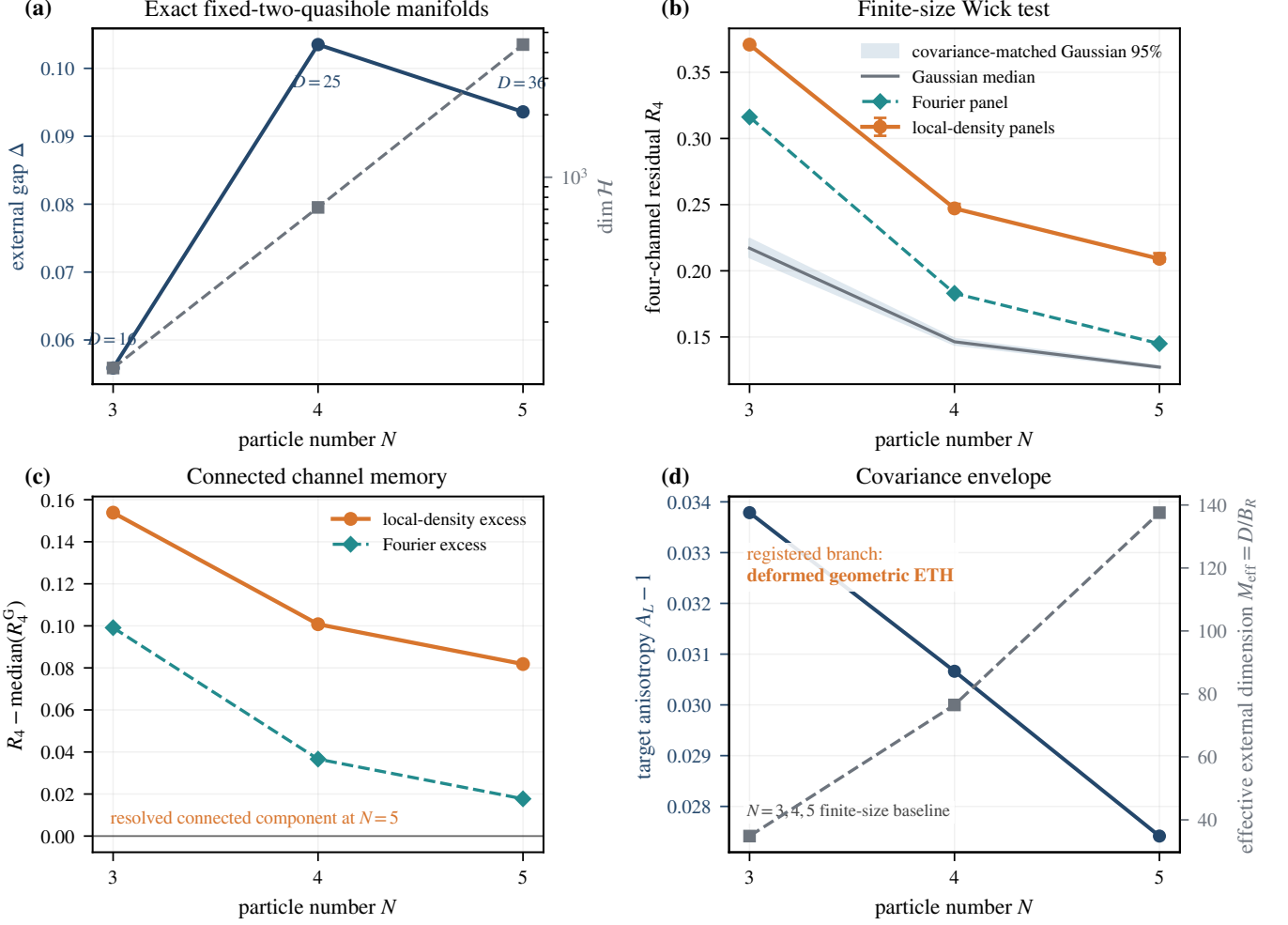


FIG. 6. **Gauge-invariant four-channel Wick test on a genuine many-body sequence.** (a) Fixed-two-quasihole manifolds $(N, n_\phi, D) = (3, 8, 16), (4, 10, 25), (5, 12, 36)$, including the external gap and full Hilbert-space dimension. (b) Physical local-density and structured Fourier residuals against covariance-matched finite- (D, M, m) Gaussian intervals. (c) The physical excess over the Gaussian median decreases with N and remains resolved at $N = 5$. (d) Measured target anisotropy and effective external dimension entering the two Wick coefficients. Intervals resample 24 complete operator panels or 2000 complete Gaussian tensors, with every matrix retained as one observation.

matched Gaussian median decreases from 0.21708 to 0.14638 to 0.12723. At the largest size the physical excess is 0.08183, resolved relative to the Gaussian interval $[0.12657, 0.12797]$.

The four-channel residual decreases along the genuine many-body sequence and retains a resolved connected component at the largest size. This selects the registered branch `deformed_geometric_eth`: progressive Gaussianization with a resolved connected operator-channel component. The $N = 3, 4, 5$ sequence supplies the finite-size baseline for an $N = 6$ scaling test.

B. A closed surface at fixed spectrum and Chern class

For the topology test, the parameter space is the complete twist torus T_{θ_x, θ_y}^2 . Starting from the physical projected-contact parent, define

$$\begin{aligned} \mathcal{U}_g(\theta) &= e^{ig \sin \theta_x G_x} e^{ig \sin \theta_y G_y}, \\ H_g(\theta) &= \mathcal{U}_g(\theta) H_0(\theta) \mathcal{U}_g^\dagger(\theta), \end{aligned} \quad (35)$$

where G_x, G_y are deterministic nearest-neighbor Hermitian one-body generators. The bosonic lift is globally periodic. Hence the complete spectrum and gap are identical for every g , and $\mathcal{U}_g$ is a bundle isomorphism that preserves C_1 . Nevertheless the connection acquires the projected one-form in Eq. (B13), so non-Abelian Wilson holonomy may change.

Polar-unitary links and plaquette products implement the lattice connection [23]. We retain determinant topology separately from relative Wilson eigenphases. The circular adjacent-gap ratio is branch free, and the Wilson form factor

$$K_W(k) = \frac{1}{D} |\text{Tr } W^k|^2 \quad (36)$$

is invariant under a common phase. Both are compared with 10000 independently sampled CUE matrices at the same D . Random local $U(D)$ frame gauges provide dedicated invariance tests and remain separate from physical scrambling.

The accepted 16×16 and 20×20 meshes give $C_1 = 6$ for $N = 3$ and $C_1 = 10$ for $N = 4$, with external gaps 0.051741 and 0.094695. The minimum accepted determinant-branch margin is 1.17980, the minimum neighboring-overlap singular value is 0.83693, and the maximum random-gauge Wilson-phase error is $6.85e - 15$. A retained 12×12 preflight identifies a branch alias for one $N = 4, g = 1$ seed and motivates the converged $16/20$ mesh pair.

Under noncommuting generators, the seed-cluster confidence interval for the change in $\langle r_W \rangle$ is $[0.02777, 0.05068]$ at $N = 3$ and $[0.01805, 0.02835]$ at $N = 4$. Both intervals are strictly positive. At $g = 1$, the medians 0.32115 and 0.33175 occupy a structured range below the CUE medians 0.60153 and 0.60079, accompanied by a distinct Wilson SFF.

The Chern class and complete spectrum remain fixed while Wilson statistics change significantly and occupy a structured class distinct from the circular-unitary reference. The selected branch is therefore **fixed_chern_deformed_holonomy**: fixed determinant-bundle topology coexists with tunably deformed, structured non-Abelian holonomy. Together with Fig. 6, it identifies two complementary memories beyond curvature eigenvalue repulsion: a resolved four-channel cumulant and a structured Wilson sector distinct from CUE.

VIII. DISCUSSION

The central result can be stated directly:

Exact degeneracy makes the energy spectrum silent, while non-Abelian projector geometry reveals a random-matrix correlation ramp, a measurable matrix-element cumulant, and deformable global holonomy inside one fixed Chern class.

Each clause has a dedicated test. Equation (1) proves spectral flatness. The finite-Jacobi kernel supplies the geometric reference curve. The full-rank Fourier family resolves structured and random local tangents at equal active rank. The two blockwise interventions establish independent spectral and geometric axes. The four-channel residual promotes the analysis from curvature eigenvalues

to invariant response maps. The closed twist torus separates determinant topology from relative Wilson transport while preserving every energy.

The resulting picture is sharper than the statement that Berry curvature is a random matrix. The physical one-point density is covariance deformed, long-range counting fluctuations retain microscopic information, and exact boundary atoms can occupy a macroscopic fraction of the spectrum. Local finite-Jacobi universality, progressive response-channel Gaussianization, and structured global holonomy therefore form three ordered geometric scales. The analytic atom theorem adds a fourth scale by separating deterministic geometric modes from the correlated continuous sector.

The relation to Chen *et al.* [13] is direct at the level of the diagnostic and complementary at the level of mechanism. Their proposal addresses exactly degenerate BPS sectors, supersymmetric black-hole microstates, smooth horizonless geometries, and Chern topology over the $\mathcal{N} = 2$ SYK moduli space. Our frustration-free condensed-matter mechanism adds the following capabilities:

1. a parameter-free finite- D Jacobi prediction for metric-normalized curvature;
2. a same-parent, same-rank structured tangent reference;
3. causal separation of PHP from $P(\partial H)Q$;
4. an exact boundary-atom correction to the connected plateau;
5. a gauge-invariant four-channel test on a genuine $N = 3, 4, 5$ sequence; and
6. a closed-surface separation of fixed Chern topology and deformable Wilson holonomy.

The common principle is that chaos can be encoded in how a degenerate state bundle moves over coupling space even when every state shares one energy.

The current calculation establishes a precise finite-size scope. The fixed-rank sequence quantifies curvature scaling through $D = 800$. The fixed-projector Poisson-to-GUE path is a controlled block intervention. The curvature form factor is treated as a geometric Fourier scale. The many-body response sequence reaches $N = 5$, where the connected four-channel component remains resolved. The Wilson family is an exact isospectral bundle construction whose final statistics define a structured class distinct from CUE. These statements provide clean starting points for larger systems, local microscopic deformations, and dynamical comparisons.

The next analytic target is the observed decrease of R_4 . An extension through $N = 6$, combined with a locality-based derivation, can distinguish an emergent Wick law from a deformed fixed tensor. A second exact-degeneracy mechanism can establish cross-model universality. On the topological side, microscopic local deformations that

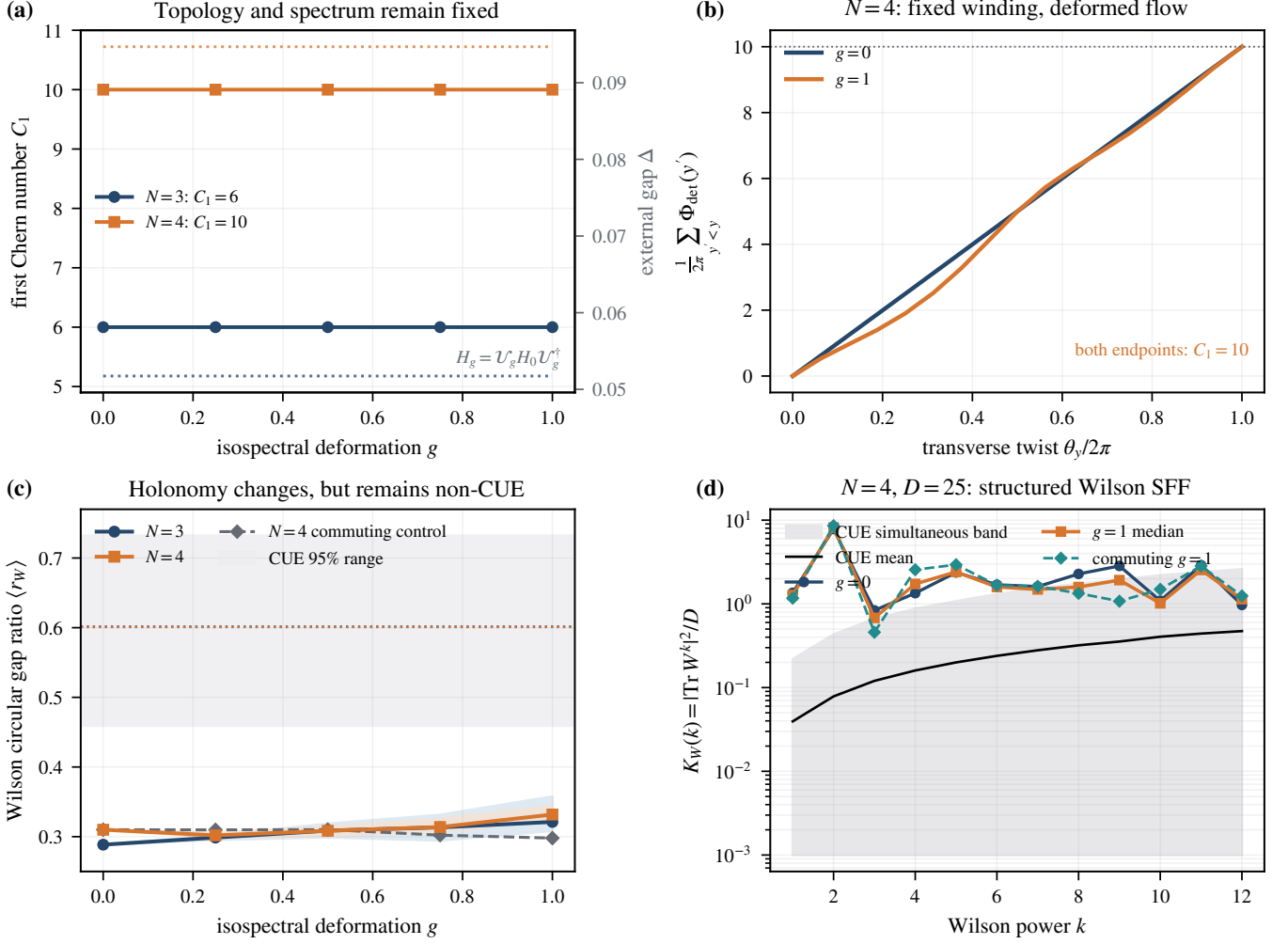


FIG. 7. **Fixed Chern topology with deformed non-Abelian Wilson holonomy.** (a) The periodic ambient orbit leaves $C_1 = 6, 10$ and the external gaps fixed for $N = 3, 4$. (b) For $N = 4$, accumulated branch-resolved determinant flux has winding ten at both endpoints, while its local distribution over the transverse twist changes. (c) Seed-cluster intervals of the Wilson circular gap ratio move significantly with g and occupy a structured range distinct from CUE; a commuting-generator path supplies the reference. (d) The $N = 4$ Wilson SFF at $g = 0$, $g = 1$, and the commuting endpoint compared with the independently sampled CUE mean and simultaneous band. Eight complete noncommuting generator seeds are resampled as clusters.

preserve an exact kernel can test the fixed- C_1 , structured-Wilson branch beyond the controlled ambient orbit. A weak lifted-band or driven calculation can then connect the geometric hierarchy to real-time dynamics.

For exactly degenerate quantum matter, the practical prescription is clear. Compute the isolated projector response, whiten curvature by its quantum metric, compare the continuous spectrum with the exact finite-rank Jacobi process, test invariant higher response cumulants, integrate determinant curvature on a closed surface, retain the relative Wilson spectrum, and map the scale at which each reference becomes active. Under this protocol, degeneracy becomes a resource: it isolates a geometric channel of quantum chaos with local, operator-level, and topological resolution.

Appendix A: Numerical methods, inference, and reproducibility

1. Physical parent and response cache

The single-particle Kapit–Mueller hopping matrix is constructed with magneto-periodic boundary conditions and projected to its flat lowest band [16]. The many-body contact interaction is represented as a positive constraint factor before diagonalization. Its numerical null space supplies P ; the nonzero eigenvectors and energy denominators supply R in Eq. (3). For a site-potential coefficient vector v , linearity is used to cache $X[v] = PV[v]R$. Every sampled channel is checked against a direct many-body Hilbert–Schmidt norm, and every normalized cur-

vature is audited for $YY^\dagger = I_r$, Hermiticity, rank, and $\text{spec } \Omega \subset [-1, 1]$.

The random-local ensemble has 20000 tangent pairs in eight independently seeded blocks. A fixed permutation determines a 12000/4000/4000 training/validation/test split before any density, covariance, or form-factor comparison. The 4000-matrix test set is used in all v2 physical figures. The Fourier reference enumerates all nonzero momenta and combines them into the 12 distinct $\mathbf{q} \leftrightarrow -\mathbf{q}$ orbit-level statistical units.

2. Exact finite-Jacobi quadrature

Equation (14) is evaluated directly by quadrature. At Gauss–Jacobi nodes x_i and weights ω_i of order $q = 512$, we form the weighted orthonormal basis

$$\Phi_{in} = \sqrt{\omega_i} \frac{P_n^{(a,a)}(x_i)}{\sqrt{h_n^{(a)}}}, \quad 0 \leq n < k, \quad (\text{A1})$$

where

$$h_n^{(a)} = \frac{2^{2a+1}}{2n+2a+1} \frac{\Gamma(n+a+1)^2}{\Gamma(n+1)\Gamma(n+2a+1)}. \quad (\text{A2})$$

The discrete projection kernel is $\mathcal{P} = \Phi\Phi^\dagger$, its diagonal is the quadrature mass of ρ_k , and the midpoint cumulative sum of that diagonal implements Eq. (12). The registered $D = 50, M = 170$ calculation has density-mass error 4.19×10^{-13} and basis-orthogonality error 4.18×10^{-13} . Repeating the complete curve at quadrature orders 384 and 512 changes it by less than 5×10^{-5} .

As an independent test, two complex Wishart matrices generate the matrix-beta Jacobi ensemble. For rank 16, 1500 independently sampled spectra agree with the determinantal-kernel connected SFF within 0.075 over the registered test grid. For $r > M$, exact ± 1 atoms are appended algebraically. The identity in Eq. (17) is satisfied to floating-point zero, and the raw decomposition Eq. (19) closes within 1.14×10^{-13} .

3. Interpolation ensembles

The geometric interpolation uses

$$g = 0, 0.02, 0.05, 0.10, 0.20, 0.40, 0.70, 1.00.$$

At every positive g , 4000 independent random local residual pairs are mixed with a registered Fourier pair through Eq. (29). All 28000 positive- g curvature matrices have active rank 50. The complete script-generated scalar adjacent-gap intervals are retained with the numerical artifacts; the Haar–Jacobi simultaneous interval is 0.598440–0.600349.

The spectral interpolation uses

$$\alpha = 0, 0.10, 0.20, 0.35, 0.50, 0.70, 0.85, 1.00$$

and 4000 matrices per value. For every sample, the same Poisson and GUE endpoints are used across α , which reduces interpolation noise while different samples remain independent. The target projector is recomputed numerically from the embedded full Hamiltonian rather than assumed fixed. The maximum measured distance in Eq. (27) and the zero curvature-spectrum error are hard delivery gates.

4. Connected estimators and simultaneous inference

Raw, disconnected, and connected form factors are retained separately for every ensemble:

$$\hat{K}_{\text{raw}} = \frac{1}{DN_s} \sum_{s=1}^{N_s} |Z_s|^2, \quad \hat{K}_{\text{disc}} = \frac{1}{D} \left| \frac{1}{N_s} \sum_{s=1}^{N_s} Z_s \right|^2. \quad (\text{A3})$$

Their difference is the displayed connected estimator. This common convention is used for the exact-degenerate energy baseline, physical curvature, Fourier control, interpolation ensembles, and Monte Carlo Jacobi reference. At $\tau = 0$, connected values below 5×10^{-15} are set to zero only after raw-minus-disconnected subtraction.

Complete matrices are the independent statistical units. Physical curves are first averaged within eight independent seed blocks; Fourier curves are averaged within the twelve momentum orbits; the Haar reference uses complete matrices. We construct 10000-replicate Gaussian-multiplier or hierarchical-block processes and take the 95th percentile of the maximum studentized deviation over each registered grid. Thus every shaded region used for an onset or compatibility statement is a simultaneous confidence band across the complete registered grid.

For an SFF residual band $[\ell(\tau), u(\tau)]$, the registered compatibility onset is the first grid point $\tau_* \in [0.25, 1.5]$ such that $0 \in [\ell(\tau), u(\tau)]$ for every subsequent registered point. An interpolation value g is said to possess a non-trivial compatibility window only if at least eight grid points remain. The number-variance compatibility extent is the largest L in the contiguous initial interval beginning at $L = 0.25$ whose residual band contains zero. These definitions were fixed before reading the reported onsets.

5. One-point covariance and finite-rank sequence

The v1 analysis retained in the present expanded article tests a distinct observable: the ensemble-mean ambient row-space projector

$$\bar{\Pi} = \mathbb{E}[Y^\dagger Y]. \quad (\text{A4})$$

A covariance-deformed Gaussian row-space ensemble learned from 1024 training matrices and selected on validation data reduces the held-out density L^1 error from 0.2910 to 0.1237, while preserving the local gap ratio. The exact Jacobi SFF in v2 remains parameter free; this

result explains why one-point and long-range observables can retain structure alongside local two-level agreement.

For the rank sequence, cyclic $(1, 2)$ -admissible root configurations and their one-body descendants are generated independently of the lattice cache. The seven registered sample counts are

$$2000, 2000, 2000, 1000, 1000, 500, 250.$$

Atoms are labeled by Eq. (16) before their numerical eigenvalues are inspected. The final $D = 800$ case therefore contains $120 + 120$ exact atoms and 560 interior levels in every matrix. Because $N = 3$ remains fixed, changes along this sequence are described as finite-rank evidence only.

6. Many-body Wick sequence

The fixed-two-quasihole sequence uses $n_\phi = 2N + 2$ and the exact cyclic zero-mode ranks $D = 16, 25, 36$. The $N = 3, 4$ parents are diagonalized densely. For $N = 5$, the complete 36-column kernel frame is obtained by a block sparse eigensolver; its residual norm is 3.59×10^{-9} . Complement resolvents are evaluated at two positive shifts and extrapolated linearly to zero shift. The maximum $N = 5$ relative equation residual is 2.75×10^{-7} , and the maximum two-shift difference is 3.94×10^{-4} . The physical sequence begins at $N = 3$, since the $N = 2, n_\phi = 6$ Kapit-Mueller parent has 12 numerical zero modes compared with the cyclic target rank nine.

For each size, 24 deterministic balanced local-density panels supply the physical resampling units. Each panel has eight operators. The finite-size null uses 2000 complete covariance-matched Gaussian response tensors per size. All target and external gauge checks are applied before the four-channel tensor is evaluated.

7. Closed twist bundle and Wilson statistics

The base twist bundles are diagonalized at every point of the periodic mesh. Physical many-body overlaps include the exact symmetric bosonic lift of the single-particle orbital overlap. The undeformed 6×6 and 8×8 meshes establish the base integer. A 12×12 preflight identifies a determinant-branch alias and a shifted $C_1 = 8$ for one $N = 4, g = 1$ seed. The accepted production pair 16×16 and 20×20 resolves the full $g \leq 1$ orbit, with the coarse alias artifact retained for provenance.

Eight independently seeded noncommuting generator pairs and one commuting control are evaluated at $g = 0, 0.25, 0.5, 0.75, 1$. Generator seeds are the cluster units in the bootstrap. The CUE reference contains 10000 independent matrices per rank. Per-loop arrays are retained within their generator-seed clusters for bootstrap inference.

8. Reproducibility gates

The task-local delivery requires regenerated hashes for every source artifact; exact registered sample counts, ranks, and interpolation points; disjoint training, validation, and test indices; matrix-level confidence calculations; finite-Jacobi mass, orthogonality, Monte Carlo, atom, and raw-closure checks; fixed-projector error below 10^{-12} ; synchronized figure hashes; resolved LaTeX references; and a render record for every compiled page. Task isolation keeps the v3 implementation fully self-contained. Additional gates require both independent v3 audits, the registered 16/20 topology pair, the retained coarse-mesh alias preflight, exact result branches, executable Wick and Chern identities, and source hashes for Figs. 6 and 7.

Appendix B: Gauge-invariant response cumulants and bundle topology

1. Response maps and gauge covariance

Let $\Phi_P(\lambda)$ and $\Phi_Q(\lambda)$ be orthonormal frames for the D -dimensional zero-mode fiber and its resolved complement, respectively. For a fixed local perturbation V_μ , we use the complement-to-fiber response map

$$X_\mu = R_Q Q V_\mu P, \quad R_Q = [Q(H - E_0)Q]^{-1}, \quad (B1)$$

whose matrix in these frames has shape $M \times D$. Independent frame changes $\Phi_P \mapsto \Phi_P U_P$ and $\Phi_Q \mapsto \Phi_Q U_Q$ act as

$$X_\mu \mapsto U_Q^\dagger X_\mu U_P. \quad (B2)$$

Consequently the channel-label covariance

$$C_{\mu\nu} = \frac{1}{D} \text{Tr}(X_\mu^\dagger X_\nu) \quad (B3)$$

is independent of both frame gauges. On the strictly positive support of C , define $\tilde{X}_\mu = (C^{-1/2})_{\mu\nu} X_\nu$. The algebraic support projection retains precisely the accessible perturbation directions. The normalized four-channel tensor

$$T_{\mu\nu\rho\sigma} = \frac{1}{D} \text{Tr}(\tilde{X}_\mu^\dagger \tilde{X}_\nu \tilde{X}_\rho^\dagger \tilde{X}_\sigma) \quad (B4)$$

is therefore invariant under $U(M) \times U(D)$ frame rotations. Label-space rotations transform all four indices covariantly, so the Frobenius norm of its connected part is label-basis independent as well.

2. Separable Gaussian contraction

The covariance-matched null model is a complex Gaussian response tensor satisfying

$$\mathbb{E}[(X_\mu)_{ai}(X_\nu)_{bj}^*] = \delta_{\mu\nu} R_{ab} L_{ij}, \quad \frac{\text{Tr } R \text{ Tr } L}{D} = 1. \quad (B5)$$

TABLE I. Registered matrix-element sequence. The Gaussian column is the covariance-matched finite-size median; “excess” is the physical median minus that value.

N	n_ϕ	D	$\dim \mathcal{H}$	Δ	R_4	R_4^G	excess
3	8	16	120	0.055872	0.37093	0.21708	0.15385
4	10	25	715	0.103502	0.24715	0.14638	0.10078
5	12	36	4368	0.093593	0.20906	0.12723	0.08183

TABLE II. Accepted closed-surface topology and Wilson statistics. The final and CUE columns are median circular adjacent-gap ratios.

N	D	C_1	$\Delta_{\min}$	branch margin	overlap minimum	$\langle r_W \rangle_{g=1}$	$\langle r_W \rangle_{\text{CUE}}$
3	16	6	0.051741	2.06643	0.85325	0.32115	0.60153
4	25	10	0.094695	1.17980	0.83693	0.33175	0.60079

Here R and L are positive external- and target-space covariances; the last identity is precisely the label-whitening normalization. Expanding Eq. (B4) in indices gives

$$T_{\mu\nu\rho\sigma} = \frac{1}{D} \sum_{abij} X_{\mu,ai}^* X_{\nu,aj} X_{\rho,bj}^* X_{\sigma,bi}. \quad (\text{B6})$$

Complex Wick contraction permits two pairings. Pairing (μ, ν) and (ρ, σ) yields

$$\frac{(\text{Tr } R)^2 \text{Tr } L^2}{D} \delta_{\mu\nu} \delta_{\rho\sigma} = A_L \delta_{\mu\nu} \delta_{\rho\sigma}, \quad A_L = D \frac{\text{Tr } L^2}{(\text{Tr } L)^2}. \quad (\text{B7})$$

Pairing (μ, σ) and (ρ, ν) instead yields

$$\frac{\text{Tr } R^2 (\text{Tr } L)^2}{D} \delta_{\mu\sigma} \delta_{\rho\nu} = B_R \delta_{\mu\sigma} \delta_{\rho\nu}, \quad B_R = D \frac{\text{Tr } R^2}{(\text{Tr } R)^2}. \quad (\text{B8})$$

Thus

$$T_{\mu\nu\rho\sigma}^{\text{Wick}} = A_L \delta_{\mu\nu} \delta_{\rho\sigma} + B_R \delta_{\mu\sigma} \delta_{\rho\nu}, \quad R_4 = \frac{\|T - T^{\text{Wick}}\|_F}{\|T^{\text{Wick}}\|_F}. \quad (\text{B9})$$

This defines a finite-dimensional distribution of R_4 . The comparison ensemble is generated with the measured spectra of L and R , passed through the identical whitening map, and evaluated at the same (D, M, m) .

3. External covariance through a compact Gram space

Stack the whitened maps as $\mathbf{X} = (\tilde{X}_1 \tilde{X}_2 \cdots \tilde{X}_m)$, an $M \times mD$ matrix. The external covariance and its Gram reduction are

$$R = \frac{1}{m} \mathbf{X} \mathbf{X}^\dagger, \quad G = \frac{1}{m} \mathbf{X}^\dagger \mathbf{X}. \quad (\text{B10})$$

If $Ru = \lambda u$ with $\lambda \neq 0$, then $\mathbf{X}^\dagger u \neq 0$ and $G(\mathbf{X}^\dagger u) = \lambda(\mathbf{X}^\dagger u)$; the reverse implication follows by applying $\mathbf{X}$.

Hence R and G have identical nonzero spectra, including multiplicities. The coefficient B_R is therefore evaluated efficiently in the mD -dimensional Gram space.

4. Periodic ambient conjugation

Let $E_0 \rightarrow T^2$ be the base zero-mode bundle and let $\mathcal{U}_g : T^2 \rightarrow U(\mathcal{H})$ be smooth and periodic. The deformed Hamiltonian and projector are

$$\begin{aligned} H_g(\theta) &= \mathcal{U}_g(\theta) H_0(\theta) \mathcal{U}_g^\dagger(\theta), \\ P_g(\theta) &= \mathcal{U}_g(\theta) P_0(\theta) \mathcal{U}_g^\dagger(\theta). \end{aligned} \quad (\text{B11})$$

Every eigenvalue, the kernel dimension, and the external gap are identical pointwise. More strongly, $(\theta, v) \mapsto (\theta, \mathcal{U}_g(\theta)v)$ is a globally defined vector-bundle isomorphism $E_0 \simeq E_g$. Naturality of Chern classes therefore gives

$$c_1(E_g) = c_1(E_0). \quad (\text{B12})$$

The same conclusion follows at the level of connections. For a local frame $\Phi_g = \mathcal{U}_g \Phi_0$ and the convention $A = i\Phi^\dagger d\Phi$,

$$A_g = A_0 + i\Phi_0^\dagger \mathcal{U}_g^\dagger (d\mathcal{U}_g) \Phi_0 \equiv A_0 + \mathcal{A}_g. \quad (\text{B13})$$

The difference $\mathcal{A}_g$ is a globally defined endomorphism-valued one-form on the identified bundle. Since $\text{Tr}[A_\mu, A_\nu] = 0$,

$$\text{Tr } F_g - \text{Tr } F_0 = d \text{Tr } \mathcal{A}_g, \quad \frac{1}{2\pi} \int_{T^2} (\text{Tr } F_g - \text{Tr } F_0) = 0. \quad (\text{B14})$$

Equation (B13) also explains the compatibility of fixed topology and variable holonomy. For a generic projected one-form $\mathcal{A}_g$, the path-ordered exponential changes around a noncontractible cycle. The determinant winding remains constrained by Eq. (B12), while the relative $SU(D)$ eigenphases may deform.

5. Discrete links and branch control

On a periodic mesh, neighboring physical many-body frames have overlap M_ℓ . We use its polar unitary $U_\ell = M_\ell(M_\ell^\dagger M_\ell)^{-1/2}$, so arbitrary local $U(D)$ frame gauges act on links by endpoint conjugation. Plaquette products transform by similarity, and both their eigenphases and $\arg \det W_\square$ are gauge invariant. The lattice first Chern estimator is

$$C_1 = \frac{1}{2\pi} \sum_{\square} \arg \det W_\square. \quad (\text{B15})$$

We require positive overlap singular-value and determinant-branch margins, agreement of determinant and trace-log estimators, and the same integer on two meshes. Directly unwrapping determinant Wilson phases can alias when the flux accumulated across an entire transverse strip exceeds π ; Fig. 7 instead accumulates the branch-resolved plaquette flux within each strip. The retained 12×12 alias preflight and accepted $16 \times 16/20 \times 20$ pair make this distinction explicit.

CODE AND DATA AVAILABILITY

The public repository <https://github.com/JunkaiWang-TheoPhy/Chaos-of-Quantum-Geometry>

contains the complete source, tests, seven figures, generated tables, compact numerical artifacts, and the audited article. From `01.task_folder/task_05/script`, `bash run.quick.verify.v1.sh` verifies the release contract and focused analytic tests, while `bash run.full.recompute.v1.sh` executes the complete production pipeline. The machine-readable `output/release_manifest.v1.json` records every Git-resident artifact and every production array with its path, byte size, SHA-256 hash, storage class, and producing command.

ACKNOWLEDGMENTS

The numerical artifacts, fixed train-validation-test split, matrix-level uncertainty estimates, and figure-generation scripts are retained with the source of this manuscript.

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